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Consumer perceived risk: conceptualisations and models

Consumer
perceived risk

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Keywords *Consumer behaviour, Consumer's risk, Marketing strategy, Model, Perception, Risk*

Received August 1993

Revised November 1995

January 1998

Abstract *Reviews the literature on consumer-perceived risk over the past 30 years. The review begins by establishing perceived risk's relationship with related marketing constructs such as involvement and trust. It then tackles some debates within the literature, concerning subjective and objective risk and differences between the concepts of risk and uncertainty. It describes how different models have been devised and operationalised to measure risk and how these have developed over the years. Aims to identify and report the theoretical and model developments over the past 30 years and to propose criteria which researchers can use in deciding the most useful model for their own research. The criteria are: understanding, prediction, suitability for reliability and validity assessment, practicality and usability. It is suggested that the basic two-component model is still the most generally useful for researchers and practitioners alike.*

Introduction

In 1960, when Bauer first brought the concept of risk to the attention of the American marketing community he stated that:

I have neither confidence nor anxiety that my proposal will cause any major stir. At most, it is to be hoped that it will attract the attention of a few researchers and practitioners and at least survive through infancy (Bauer, 1960, p. 389).

More than 30 years later, the perceived risk concept has come through infancy to adulthood and has established a tradition of research unparalleled in consumer behaviour research. Perceived risk continues to receive attention from both practitioners (Farquhar, 1994) and academics (Grewal *et al.*, 1994) and has been applied in a wide range of areas including intercultural comparisons (Alden *et al.*, 1994), food technology (Frewer *et al.*, 1994), dental services (Coleman *et al.*, 1994), banking (Ho and Victor, 1994) and apparel catalogue shopping (Jasper and Ouellette, 1994). So, why do marketing practitioners and researchers continue to be interested? First, perceived risk theory has intuitive appeal and plays a role in facilitating marketers seeing the world through their customer's eyes. Second, it can be almost universally applied and its versatility has been demonstrated in a wide range of applications, from spaghetti (Cunningham, 1967) to industrial reprographic equipment (Newall, 1977). Third, it is suggested that perceived risk is more powerful at explaining consumers' behaviour since consumers are more often motivated to avoid mistakes than to maximise utility in purchasing. Fourth, risk analysis can be used in marketing resource allocation decisions. For example, a study of risk relievers used by consumers can help to increase marketing efficiency by channelling resources into strategies which consumers find more useful, while withdrawing them from those which they find less useful. Risk perception

European Journal of Marketing,
Vol. 33 No. 1/2, 1999, pp. 163-195,
© MCB University Press, 0309-0566

analysis can also be helpful in brand-image development, targeting, positioning and segmentation; e.g. by highlighting risk aspects in comparative advertising; repositioning commodity products to give added value, and segmenting consumers as on the basis of their risk-reducing strategy usage. Finally, examining risk perceptions can generate new product ideas. In a recent study of breakfast cereals, one of the risks consumers perceived was a result of disliking milk. This suggested the development of non-milk-based breakfast products such as Kellogg's Pop Tarts (Mitchell and Boustani, 1993).

However, for researchers new to perceived risk there appears an overwhelming number of papers reporting empirical findings and theoretical developments to review before beginning any research. This paper attempts to summarise some of the major developments in model building and testing over the past 30 years and give some guidelines to assess the usefulness of the various conceptualizations. It suggests a set of criteria for assessing the models and uses it to comment on some of the main models found in the literature. In doing so, consideration is given to how the conceptualization of risk has changed in the course of the literature's development from the time risk taking was conceived, in 1960, to 1997. Areas for further exploration and research are identified as is the philosophical perspective of the perceived risk literature and the concept's relationship to involvement and trust.

Perceived risk debates and definitions

This section discusses several important ideas within the literature including the relationship between objective and subjective risk, the difference between risk and uncertainty, inherent and handled risk as well as introducing various authors' ideas on how risk can be characterised and defined.

Objective and subjective risk

One of the first debates met in the literature is that on the existence of objective risk. Bauer (1960) strongly emphasised that he was concerned only with subjective (perceived) risk and not "real world" (objective) risk. Unlike actuaries or accountants, who may have vast amounts of accurate historical data to estimate the risk of occurrences, the average consumer has limited information, a reduced number of trials to consider and a semi-reliable memory. In many instances, consumers will be faced with a completely new purchase which they have never encountered before. This makes accurate assessment of risk almost impossible. Even if the consumer could calculate accurately the risk involved, it is not this objective risk which motivates behaviour, but the consumer's subjective impressions of it. Any measurement of risk perception must be developed with these limitations in mind.

Stone and Winter (1985) argue that there is no such thing as objective risk; except perhaps for physical risk. They believe that it is impossible to have some real world or objective social, psychological, time, financial and performance risk. The consistency of their argument is broken when they accept a doctor as

being an “objective” risk assessor for physical risk, but reject a financial expert as being able to give an objective assessment on financial risk.

The question of objective versus subjective risk raises the issue of a researcher’s philosophical perspective. Unlike many subjects which divide researchers along the lines of how they view the world, perceived risk encourages a convergence of these divergent views. A tenet of scientific realism is that the world exists independently of it being perceived. Scientific realists, therefore, would attempt to search for the objective risk within any given situation. The relativist perceived risk researcher would not accept the existence of an objective risk, arguing that risk is relative to the perceiver and nothing can be measured beyond that. While this is a fundamental point of difference, the two schools of thought, in practice, are unified by the need to measure subjective risk, namely that risk which is perceived by the consumer and which motivates behaviour (see Figure 1). For realists to concede that the subjective impressions of an observable phenomenon are worth conceptualising and measuring is a major bridge of the philosophical divide. Equally relativists would seem happy to concede to use the scientific tools of the realist to analyse risk, philosophically secure in the knowledge that it is an individual and relativist perspective which is attempted to be measured.

The author believes that objective risk must exist in theory. What is lacking is the ability to measure it. Time, money and, to some degree, physical harm can be measured by experts using specific measurement tools. Psychosocial risk is less easily calculated. Although psychometric scales, in some cases, could be devised to measure such phenomena, the risk is so complex and potentially changeable, that it is difficult to measure accurately. An objective measure of risk is therefore difficult to obtain, but that is not to say that it does not exist. All that can be easily measured is the subjective or perceived risk.

Cooper *et al.* (1988) provide evidence for the necessity to differentiate between differences in risk perception and risk attitude. They found the main differentiation between entrepreneurs and other managers was not a greater preference for risk, but an overly optimistic perception of the risks involved. Knowing which is more important is useful, since if changes in risk perception are the driving force, then an effective remedy should be to target cognitive processes with information aimed at a more realistic risk perception; while if

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	Relativism	Positivism
Subjective risk	The only risk which exists and that can be measured	Willing to accept its existence and the need to measure it
Objective risk	Not willing to admit exists	Attempts should be directed at conceptualising and measuring this where possible

Figure 1.
Philosophical beliefs
about perceived risk

risk preferences are more influential than intervention needs to focus on consumers' emotional responses (Weber and Milliman, 1997).

Risk and uncertainty

Extending this debate leads us naturally on to the difference between risk and uncertainty. Knight's (1948) definition separates the concepts of risk and uncertainty. Knight proposed that "risk" has a known probability while "uncertainty" exists when knowledge of a precise probability is lacking. Even though this distinction between uncertainty and risk has been drawn in terms of distribution of outcomes, invariably marketers have allowed the two concepts to be used synonymously. This is most probably because marketers feel that consumers never really know the exact probability of an outcome. According to Knight, therefore, we should be talking about perceived uncertainty. But what if the consumer can place a subjective probability on an event? It may have no relation to the objective probability involved in the situation, but it is still a "known" probability. Does this allow marketers to use Knight's definition of risk? Historically, the term risk has been consistently used, although this particular argument for so doing has not been articulated.

Cunningham (1967, p. 83) makes the point that uncertainty or consequences may involve either a known or unknown probability. He suggests that it makes little difference, in his conceptualization, whether the consumer knows that there is an 80 per cent chance that he will make a bad purchase or whether the consumer thinks that he just "might" make a bad purchase. In addition, he argues intuitively that known probabilities are extremely rare in purchase behaviour, and that even when they are available, the consumer is unlikely to think in terms of them. Therefore, the concept of perceived risk used by consumer researchers bears a closer relationship to the concept of partial ignorance (where neither the consequences of alternatives nor their probabilities of occurrence are accurately known).

Although some authors have implied that "dealing with information implies the handling of uncertainty. ... in a word, it means to handle 'risk'" (e.g. Nicosia, 1969, p. 162), Peter and Ryan (1976) take the position that risk and uncertainty are clearly not the same. They suggest that to equate perceived risk to uncertainty adds little in terms of meaning specification because if perceived risk was equivalent to the concept of uncertainty and a consumer was perfectly certain that a brand is totally unacceptable for purchase there would be no uncertainty or perceived risk, by definition. However, if there is no uncertainty or perceived risk in this situation, why is the brand totally unacceptable? Risk and uncertainty must therefore be understood to be distinct from one another.

Inherent and handled risk

Inherent risk is the latent risk a product class holds for a consumer, while handled risk is the amount of conflict the product class is able to arouse when the buyer chooses a brand from a product class in his/her usual buying situation (Bettman, 1973, p. 184). Handled risk represents the end results of

information acquisition and risk-reduction processes on inherent risk. For example, a consumer may feel there is a great deal of risk associated with the product class shampoo; however, they buy their favourite brand with confidence. Dowling and Staelin (1994) have referred to this distinction as product category and product-specific risk.

Peter and Ryan (1976) suggested that the importance of loss varies by market segment and product class, but that it adds little to the explanatory power when used to weight probability of loss at the brand level. In Bettman's (1973) terms, importance of loss operates at the inherent risk level, while probability of loss operates at the handled risk level. For example, an expected financial loss of £100 per year because of poor fuel economy should be just as important to the consumer whichever car he chooses to buy. However, the probability of this loss could clearly be expected to vary from brand to brand.

Definitions

In classical decision theory, risk is most commonly conceived as reflecting variation in the distribution of possible outcomes, their likelihoods and their subjective values. Risk is measured either by non-linearities in the revealed utility from money or by the variance of the probability distribution of possible gains and losses associated with a particular alternative (Arrow, 1965; Pratt, 1964). In general, the theories assume that decision makers prefer smaller risks to larger ones, provided that other factors (e.g. expected value) are constant (Arrow, 1965).

In the consumer behaviour literature, perceived risk has been defined in many ways. Kogan and Wallach (1964) suggested that the concept of risk may have:

... two, somewhat different facets: a "chance" aspect where the focus is on probability; and a "danger" aspect where the emphasis is on severity of negative consequences.

Cunningham (1967, p. 37) conceptualised perceived risk in terms of two similar components, namely;

the amount that would be lost (i.e. that which is at stake) if the consequences of an act were not favourable, and the individual's subjective feeling of certainty that the consequences will be unfavourable.

The amount at stake is a function of the importance or magnitude of the goals to be attained, the seriousness of the penalties that might be imposed for non-attainment, and the amount of means committed to achieving the goals (Cox, 1967a, p. 38).

The definitions advanced by Kogan and Wallach (1964), Cunningham (1967) and Cox (1967a) have been criticised by some other authors. Sjöberg (1980, p. 302), for example, talks of the word risk as being well-known to be rather ambiguous and says that many more or less specific meanings have been attributed to it. He notes three broad classes of meaning: those concerned with the probability of negative events, those concerned with these negative events themselves measured in some suitable way, and those concerned with a joint

function of probability and consequences (most often their product). He suggests that perceived risk is seldom well pictured by the product of probability and consequences; something inspired by thinking in economics, "it can many times be quite misleading.... It is unfortunate, therefore, that one often finds it to be suggested as the definition of risk".

Stone and Winter (1987) view risk as an expectation of loss. The more certain one is about this expectation, the greater the risk for the individual. This understanding differs from the traditional, normative expectancy-value orientation that often views risk as "probability times pay-off", which traces its roots to the disciplines of mathematics and economics, rather than to a psychological-driven focus for risk which seems far more appropriate in consumer domains. Risk is, therefore, defined as a subjectively-determined expectation of loss; the greater the probability of this loss, the greater the risk thought to exist for an individual. One empirical study which examined three risk formulations among 80 senior business administrators found that risk conceptualized in terms of expectation of loss (R1) was not correlated with risk conceptualized in terms of uncertainty (R2 and R3 in Table I) (Stone and Winter, 1987). However, expectation of loss also showed a much stronger negative correlation with behavioural intentions and attitude than did the other two risk measures.

Vann (1984) contends that much of the confusion surrounding the study of perceived risk can be reduced by recognising several perceived distributions on each risk dimension (e.g. physical risk) rather than just one. Four perceived distributions are proposed. First, outcomes may have multiple aspects. For example, if we were to think of the performance dimension for a particular meal at a restaurant, there could be perceived outcome distributions for that meal's sweetness, saltiness, portion size, temperature, etc. (Ahtola, 1975; Sarel, 1982). The second distribution is presumed to be a distribution of the person's perceptions of the performance (quality) of the meal, as a whole, across multiple experiences. Third, would be the distribution of the perceptions of summary judgements regarding the meal (summarised across occasions). This distribution would include the perceptions of summary statements made by others as well as the person's own summary judgement. The fourth distribution is the perceived distribution of that meal as served by all restaurants within some relevant set (summarised for each restaurant) across other's and one's own summary judgements (across occasions).

This point is taken further by Vann (1985) in a later paper when he addresses how to characterise uncertainty inherent in a distribution. If a distribution is characterised by its variance or standard deviation, then its dispersion is represented but skewness and location are not. If a measure of central tendency is used to characterise the risk inherent in a distribution, the location is represented, but all the information regarding shape is not. Characterisations which utilise both the mean and the variance capture both location and dispersion information, but fail to represent skewness or minimum acceptable performance levels. Vann (1985) proposes a model which would appear to

Author	Date	Model description	Evaluation criteria ^a					Total
			Understanding	Prediction	Reliability and validity	Practicality	Usability	
Cunningham	1967	Overall risk = uncertainty × danger of consequences	3	3	3	3	3	15
Peter and Ryan	1976	Overall risk = probability of × negative consequences Brand preference in market _j = Σ_n . Probability of loss for brand _i , expected by market segment _j × consequences of poor brand choice occurring. Where n = facets of perceived risk, i = brands, j = market segments based on importance of losses	3	3	3	3	3	15
Stone and Winter	1987	Risk = expectation of loss (R1) Risk = uncertainty times consequences (R2) Risk = uncertainty towards positive and negative consequences (R3)	3	3	2	3	3	14
Deering and Jacoby	1972	CM-3 = $\frac{(4 + 5)}{2} \times \frac{(6 + 7 + 8 + 9 + 10)}{5}$ 1. How certain are you that a brand name of this product you have not tried will work as well as your present brand? 2. We all know that not all products work as well as others: compared to other products, how much danger would you say there is in trying a brand of this product that you have never used before? 3. How confident would you say you are about judging the quality of the product?	3	2	1	1	1	8

(Continued)

Author	Date	Model description	Understanding	Prediction	Reliability and validity	Practicality	Usability	Total
		4. Buying a product that gives you good results may be more important for some products listed than for others. How important would you say it is for this product to satisfy you? 5. The investment you make when you buy a product includes your time and energy, as well as money. In terms of the time, money and overall effort required to buy this product, how much would you say you invest? 6. Can most shoppers guess ahead of time how dependable this product will be if it is used repeatedly? 7. Before buying this product, can almost anyone tell how good its materials are and how well it is put together? 8. Can almost any shopper predict what the bad results will be if this product fails? 9. In general, does this product tend to fulfil your expectations? 10. Is it obvious why someone like yourself would want this product?						
Horton ^b	1976	$DR = a \times c^{ec} \times V^{ev} \times D^{ed}$ $DR =$ decision risk $C =$ sum of negative consequences: specifically the financial, performance and psychological losses associated with a poor brand choice	2	2	2	2	2	10

(Continued)

Author	Date	Model description	Evaluation criteria ^a				
			Understanding	Prediction	Reliability and validity	Practicality	Usability
Pras and Summers	1978	V = variance of importance brand attributes	3	3	2	2	2
		D = difficulty of judging brand attributes					
		a = scaling constant, ec, ev, ed = constant elasticity parameters ^a					
		$P_{ij} = \mu_{ij} + r_{ik} \sigma_{sij}$					
		P'_{ij} = the risk adjusted index for attribute _i and brand _j					
		μ_{ij} = the mean of the distribution evaluation for j on attribute i					
		r_{ik} = the consumer's tolerance for risk for attributes with respect to the range of possible ratings					
		σ_{sij} = the semi-standard deviation of the distribution (attribute i for brand _j) with respect to the mean. This will be downward semi-standard deviation if the consumer is a risk avoider and the upward semi-standard deviation if the consumer is a risk taker					
Dowling and Staelin	1994	Overall perceived risk (OPR) = product category risk (PCR) + product-specific risk (SR) Where PCR = F1 (individual level variables, attributes of the product class) and SR = F2 (purchase goals, purchase situation, specific product attribute)	3	3	3	2	2
		Notes: ^{a1} = poor; 2 = average; 3 = good; ^b The definition of variance of important product attributes was "within certain product classes brands vary greatly on attributes which are important to consumers while within other product classes, brands vary little on important attributes" (p. 696). Also D = within certain product classes it is relatively more difficult for consumers to judge which product attributes are important					

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Table 1.

overcome all the shortcomings of the other approaches to partitioning outcome distributions in modelling perceived risk. The model would: be anchored to the scale in a manner which explicitly considers minimum acceptance performance thresholds; reflect the dispersion of the outcome distribution, both at the positive and the negative ends (thereby be sensitive to both variability and to skewness); provide for trade-off decision strategies on the part of consumers, in which they could consider both possible negative consequences and possible positive consequences associated with an alternative. In addition, the negative and positive semi-variances could be weighted in a fashion which reflects the risk-seeking versus the risk-avoiding preferences of the consumer.

Although a number of other formal models have been proposed to represent decision making under uncertainty, the ones which are most widely used are those that follow from subjective expected utility theory (Bonoma and Johnston, 1979; Currim and Sarin, 1983, 1984; Hauser and Urban, 1979). In this approach, risk is modelled by reflecting the decision maker's response to uncertain outcomes defined in terms of specific probabilities of risk. Subjective expected utility theory predicts that the presence of ambiguity in probabilities should not affect how consumers make decisions; one should make the same decision whether the probability of risk is stated as 25 per cent or somewhere between 10 and 40 per cent. In these types of situations, the probabilities are ambiguous or, in effect, there is "uncertainty about uncertainty". Kahn and Sarin (1988) show that their model, which incorporates the uncertainty about probabilities, predicts different decisions for individuals who are ambiguity averse, ambiguity seeking, or ambiguity indifferent.

A relevant question, which assists our understanding of perceived risk, is where does the uncertainty come from? There are numerous sources. First, the consumer's knowledge of their own needs, purchase goals, acceptance levels and goal importance is frequently inadequate. For example, "how important is it that my car can travel at 100 mph?" (Cox, 1967d; Cox and Rich, 1964; Deering and Jacoby, 1972). Second, consumers can be uncertain about defining the range of decision alternatives, i.e. the number of suitable cars and the relative importance to the consumer of a brand's attributes is not known exactly (Pras and Summers, 1978). This uncertainty regarding what is known about the alternatives has been defined as knowledge uncertainty (Urbany *et al.*, 1989). Third, consumers may be uncertain about the predictive validity of the attributes which can be assessed beforehand, i.e. how well will they predict future performance (Cox, 1967c). Fourth, is the consumer's own perceived ability to accurately judge the outcome levels they have experienced (Barach, 1969; Bennett and Harrell, 1975; Deering and Jacoby, 1972). Cox (1967d) described this as one's "confidence value"; which is "a measure of how confident consumers are when categorising a cue as good or bad". Fifth, consumers may find it difficult to make an overall brand evaluation, i.e. is brand X better than brand Y, overall (Cox and Rich, 1964). Urbany *et al.* (1989) call this uncertainty about which alternative to choose, choice uncertainty. Finally is the potential disparity between the anticipated and the actual experience of the outcomes

(Kahnemann and Tversky, 1984), e.g. not only do preferences change over time, but also the situation within which a product will be experienced may be different to the one anticipated.

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There are numerous additional complications with theoretical concepts of risk when they are taken as descriptions of the actual process underlying choice behaviour. For example, Kunreuther (1976) suggests that individuals tend to ignore possible outcomes that are very unlikely, regardless of their consequences. There is evidence that individuals look at only a few possible outcomes rather than the whole distribution and measure variation with respect to those few points (Alderfer and Bierman, 1970; Boussard and Petit, 1967), and that they are more comfortable with verbal characterisations of risk than numerical ones. Moreover, the translation of verbal risk expressions into numerical form has shown high variability and context dependence (Budescu and Wallsten, 1985). There are suggestions that the likelihood of outcomes and their values enter into calculations of risk independently, rather than as their combined product (e.g. Slovic *et al.*, 1977). Also, different individuals may see the same risk situation in quite different ways (Kahneman and Tversky, 1982). Such ideas seem to indicate that the ways in which human decision makers define risk may differ significantly from the theoretical definitions of risk in the literature.

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Risk, involvement and trust

Perceived risk has also been found to be related to other consumer behaviour concepts, e.g. Cognitive style (Cox, 1967b). Kogan and Wallach (1964) found that self-sufficiency and independence were related positively to risk taking and rigidity was negatively related. Schaninger (1976) showed perceived risk and its components were negatively related to self-esteem, rigidity and risk taking and positively related to anxiety measures. Here, we focus on perceived risk's relationship to two important concepts, namely involvement and trust.

Risk is often viewed as an antecedent of involvement (Choffee and McLeod, 1973) particularly when the price is high and the consumer risks losing money. However, it has also been conceptualised as an intrinsic part of the involvement construct. Laurent and Kapferer's (1985) conceptualisation of involvement included four components (the product's pleasure value, its symbolic value, risk importance and probability of purchase error) of which two are related to risk. Some authors (Laaksonen, 1994) suggest that a theoretically-reasonable way to develop an intensity index for involvement is offered by the expectancy-times value structure which is very similar to some conceptualisations of risk. Much like risk attitudes, involvement has been separated into enduring and situational (Richins *et al.*, 1992). However, distinctions have also been drawn between cognitive and effective involvement (Park and Young, 1986). Like perceived risk, involvement can be at the product category or brand level.

Risk reduction is also linked to involvement as high involvement with a single brand is commonly known as brand loyalty which has been shown to be a major risk reducer (Roselius, 1971). Moorthy *et al.* (1997) produce convincing

evidence that product class involvement or low search cost are not sufficient to induce large amounts of search activity and that the existence of relative uncertainty among brands is necessary for search to be useful.

Risk is also related to the concept of trust, which has recently been given much attention in the relationship marketing literature (Berry, 1995; Dion *et al.*, 1995; Doney and Cannon, 1997; Hawes, 1994; Morgan and Hunt, 1994; Smeltzer, 1997). Ring and Van de Ven (1994) note that two views on trust can be found in the management and sociology literatures. One is a business view based on confidence or risk in the predictability of one's expectations. The other is a view based on confidence in the other's goodwill. Doney and Cannon (1997) discuss that the trust literature suggests that trusting parties must be vulnerable to some extent for trust to become operational, i.e. decision outcomes must be uncertain and important to the trustor (Moorman *et al.*, 1992; Schlenker *et al.*, 1973). In a risk-based view of trust, parties hedge against uncertain states of nature, adverse selection and ethical hazard through formal contractual means such as guarantees, insurance mechanisms and laws. Doney and Cannon (1997) provide empirical evidence to support a model which incorporates suppliers' reputation and size, their willingness to customise and keep confidential shared information and length of the relationship; all of which can also significantly affect the amount of risk in a supplier decision. Anderson and Naurus (1990, p. 45) focus on the perceived outcomes of trust and define it as "the firm's belief that another company will perform actions that will result in positive outcomes for the firm as well as not take unexpected actions that result in negative outcomes".

From these discussions, we can see that perceived risk is a necessary antecedent for trust to be operative and an outcome of trust building is a reduction in the perceived risk of the transaction or relationship. As relationships develop and trust builds, risk will decrease.

Perceived risk applications

Although perceived risk has found application in many areas, it is beyond the scope of this paper to review all of these[1]. Instead, this section outlines the main areas of application in consumer and organisational markets and draws distinctions between high and low involvement goods and between goods and services.

Food products have been a consistent feature of perceived risk studies over the years with the single most noticeable type of study being that of generic versus branded grocery items (see for example, Brooker, 1984; Chernatony, 1989; Dunn *et al.*, 1986; Toh and Heeren, 1982; Wu *et al.*, 1984). Studies which have examined a mixture of convenience and shopping goods (e.g. Deering and Jacoby, 1972; Derbaix, 1983; Hampton, 1977; Johnson and Andrews, 1971; Kaplan *et al.*, 1974; Laurent and Kapferer, 1985; Popielarz, 1967; Woodside, 1974), have shown that, in general, the higher value, more complicated and more involving products are more risky than the lower value, low-involvement simpler convenience products. The most popular products studied have been

deodorant, headache remedy, coffee, car and TV. For example, Derbaix (1983) found that for search goods characterised by highly visible attributes, psychosocial risk was more important than others. For durable, experience goods which are usually expensive, financial risk came first, and for non-durable goods physical risk was more important.

Some studies examined only complex consumer products (e.g. Asembri, 1986; Hisrich *et al.*, 1972; Lumpkin and Massey, 1983; Peter and Tarpey, 1975; Pras and Summers, 1978; Winakor *et al.*, 1980). Clothing is the most popular choice of product for study in this collection. However, the brown-goods market has largely been overlooked.

A number of authors have shown that services are riskier than products (Guseman, 1981; Lewis, 1976; Mitchell and Greatedorex, 1993). This is because of the inherent properties of services, i.e. heterogeneity, perishability, inseparability and intangibility which undermine consumer confidence and increase the perceived risk, mainly by augmenting the degree of uncertainty in the decision. The most frequently studied services are life insurance, doctors and hairdressers, then legal services, banks and dry cleaners with more recent studies focusing on professional services (e.g. Boze, 1987; Crocker, 1983; Garner and Garner, 1985; Motilla, 1983).

The versatility of perceived risk and its universal appeal for researchers keen to explain less usual consumer phenomena is demonstrated by studies on topics such as experts systems and artificial intelligence (Taunton, 1989; Wong, 1988), flexible manufacturing systems (Phillips, 1987), complaints about advertising (Lawson, 1985), financial risk assessment (Farrelly *et al.*, 1985), top executive travel (Brown, 1987) and diffusion theory (Onkvisit and Shaw, 1989).

Very many studies have examined fairly low-cost convenience food and non-food stuffs, with which consumers are little involved and within which there is minimal perceived risk. This is a problem because when risk is below a risk threshold, perceived risk theory has little explicatory power; except when these products are the subject of a consumer "scare". A priority for future research should be to use high-value products or services, for example, cars, fridges, washing machines, brown goods, boats, caravans, houses, time-share accommodation, jewellery, *objets d'art*, holidays, wedding arrangements, private pension plans, etc.

Applications in the organisational purchasing area have been fewer, but have shown that one of the main differences between organisational and consumer risk is the degree of complexity of consequences. Recent work has suggested that organisational buyers perceived not only personal and organisational risks, but also professional risks which are associated with their role as a professional within an organisation (Mitchell, 1998). Hakansson and Wootz (1979) refer to a model of organisational decision making with three dimensions of uncertainty:

- (1) need uncertainty;
- (2) transaction uncertainty;
- (3) market uncertainty,

while Valla (1982) identified five categories of risk with which a buyer must contend. These were:

- (1) technical risk;
- (2) financial risk;
- (3) delivery risk;
- (4) service risk; and
- (5) risk related to supplier/customer long-term relationships.

Proposed model assessment criteria

When examining the models and conceptualizations presented in the literature, some framework is required. Here, several criteria are proposed on which to judge the usefulness of perceived risk models. These include:

- the level of concept understanding generated;
- predictive success;
- suitability for reliability and validity testing;
- level of practical and managerial insight offered; and
- usability.

Developing any proposed model should increase our understanding of the construct or concept. Rivett (1994) notes that an important quality of a model should be its capacity to reach out into the unknown or the unknowable. He points out a model's first quality is to cut out irrelevance and to simplify.

One main objective of model building is prediction. In doing this, "we must always remember that mathematics is the vehicle which takes us to our destination and is not our destination itself" (Rivett, 1994, p. 26). In consumer behaviour research, this mainly takes the form of predicting consumers' propensity to purchase. This criterion overlaps with the validity criterion, since known-group validity can often use purchasers and non-purchasers as criterion groups to separate high and low-risk perceivers. This criterion focuses our attention on why the model is being constructed and highlights the comparison problems caused by the potential diversity of answers.

The suitability for reliability and validity assessment criterion is clearly underpinned by a positivist research paradigm. Much of the literature, while using the paradigm, has not followed through on assessing the reliability and validity of models. It is not always expected that once a model is proposed the researcher should provide all the necessary data relating to its reliability and validity in order to assess its usefulness in describing or explaining the concept, however, not to provide any data or assessment of how the model could be subject to such tests has regrettably been a common occurrence in the literature; even after 1979 when Churchill (1979) had identified assessing construct validity as being essential for developing good measures. In terms of further research, a major contribution to the marketing literature is waiting to be made from comparing the various perceived risk models and measures using known-group validity and multi-trait-multi-method matrices. Some limited

reliability and validity evidence is available (see e.g. Lumpkin and Massey, 1983); however, the quest remains a major challenge for future researchers.

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Practical implications

As noted in the introduction, the perceived risk concept can be used in numerous ways by marketing practitioners for developing risk-reducing strategies, new products and product modifications, segmentation tools and improving personal selling. While the pursuit of knowledge for knowledge's own sake is a laudable goal, in an applied discipline such as marketing, this pursuit must ultimately lead to better marketing practice. Some authors have addressed this when designing and using their models (e.g. Pras and Summer, 1978), but most have ignored this in their thinking, rightly or wrongly.

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Usability

This has an effect on understanding, because a model which is difficult to use in teaching, research or in the field is likely to be difficult to understand. In model development, some process of variable reduction has to take place. Variables can be tested for their relevance and for their sensitivity as well as their internal relationships. The variables chosen are usually related to how the model will be used, i.e. related to the objectives of the model.

The development of models and measures are rarely value free, "Measures always say something about the measurer". For example, if a man tells his friends that he has met a superb girl – 36-22-36 – it tells something about the girl (even though the units of measurement are never stated), but it tells much more about the man (Rivett, 1994, p. 20). Herein lies a problem with evaluating the literature. Some models and ideas have been designed to increase academic understanding of the perceived risk construct and therefore can make assumptions about the reader's level of understanding and awareness. These models, however, might fair poorly on understanding and usability if the audience were undergraduate students or perhaps marketing practitioners. This poses a problem. We can either evaluate the whole literature with one set of criteria in mind, e.g. the development of knowledge and understanding in its philosophical sense; or we judge each model against its own set of objectives which the researchers were trying to achieve when developing the model. The former of these approaches requires a relatively consistent opinion concerning this question of "contribution" to knowledge which is difficult to obtain. For the latter, we generally have insufficient information in order to make the observations. What we shall therefore attempt to do is highlight the main strengths and weaknesses of each model without attempting to make direct equivalent comparisons across all models. The ratings shown in Table I for the models are therefore highly judgemental and should be treated with caution.

Consumer perceived risk measurement models

Operationalizing perceived risk has resulted in many models, some of which are similar. This section considers the more useful measurement models (see Table I) beginning with the simplest and moving to the more complicated.

Basic models

Cunningham (1967) was one of the first to suggest a two-component model, with each dimension, uncertainty and dangerousness of consequence, measured on four-point scales. These were collapsed to three-point scales then combined multiplicatively to give a one-to-nine risk scale. The decision to group the perceived risk scale into three equivalent gradations was based primarily on an evaluation of the resultant cell sizes. As Cunningham himself admits, "an arbitrary method of constructing the perceived risk index was used" (Cunningham, 1967, p. 84).

This two-component model or variations on it has been the mainstay of perceived risk research over the past 30 years. A major preoccupation with researchers is deciding how the various elements of perceived risk should be combined, i.e. should the basic components of risk be multiplied or added?

In 1976, Peter and Ryan commented that the two components are usually combined multiplicatively to give an overall indication of perceived risk:

$$\text{Risk} = \text{probability of negative consequences occurring} \times \text{Importance of negative consequences}$$

Although the logic of this multiplicative model is not provided in the literature, it is likely to come from probability theory, where probabilities are multiplied by monetary value to determine the expected values of gambles (Peter and Ryan, 1976, p. 184). Peter and Ryan (1976) measured probabilities and importance of loss and correlated them with brand preference. For five out of six brands the summated perceived risk model for the high-importance segment was correlated more highly with brand preference than was the multiplicative form. They also concluded that the importance of losses may be more useful as a segmentation variable than as a component of a multiplicative perceived risk model. Most of the work in the risk area has proposed some sort of multiplicative formation (e.g. Cunningham, 1967; Sieber and Lanzetta, 1964). By contrast, Wright (1973) forcefully argues that such mathematical representations of consumer decision processes may be overly complicated.

The argument of multiplicative versus additive has continued to engage researchers over the three decades. Bettman (1973) provided evidence that an additive model fits slightly better; although the R^2 values were quite close for the two models, and the model coefficients had very similar patterns. Horton (1976) too reported that the linear model is generally superior to the multiplicative model at both the product class and aggregated levels. Finally, work by Lanzetta and Driscoll (1968) is supportive of a linear model. They suggested that a positive correlation between importance and uncertainty of consequences might lead to an additive model being better. Recent work by Joag *et al.* (1990) using a simulated industrial setting revealed that when a decision had multiple plays (e.g. purchasing 100 personal computers) decision makers combined probabilities and outcomes to form their risk perceptions in a manner consistent with a multiplicative information integration model. In contrast, when a decision had a single trial (e.g. purchasing one large mainframe

computer), information was combined in a manner consistent with a non-multiplicative integration pattern. Given the research evidence, an additive model might better predict risk perception in more cases than a multiplicative model, but the careful researcher should test both formulations.

Complex risk models

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Recently Dowling and Staelin (1994) have incorporated Bettman's inherent and handled risk notion into a formal model. They refer a person's predisposition towards a product category (inherent risk) as product-category risk, while the second component (handled risk) is referred to as product-specific risk. Acknowledgement is given to other antecedents of risk including:

- (1) levels of the attributes of a specific product, e.g. price, quality rating etc.;
- (2) the likelihood of failure that leads to negative consequences;
- (3) the individual's purchase goals;
- (4) other conditions, e.g. purchase channel.

A third major component is their concept of two types of acceptable risk related to the product category (e.g. sky diving) and a specific product within the product category. The acceptable risk level was defined as the lowest product-specific risk score associated with a subject's response that he/she would prefer to seek more information. Their study is one of the first to assess empirically the effect of an acceptance level of risk on any type of consumer behaviour. The model also incorporates risk reduction activity. For example, when product-specific risk is less than a person's acceptable risk level, the person's intention to engage in search behaviour is hypothesised not to be influenced by product-specific risk. A new method was used to assess risk by using a conjoint methodology in which the part worths (i.e. risk utilities) are estimated for each potential consequence for all product attributes for every individual.

One of the most complex measurement models has been developed by Deering and Jacoby (1972) who measured risk using ten questions (see Table I). The first composite measure (CM-1) combined responses to two questions used in previous studies (e.g. Cunningham, 1967a), questions one and two in Table I. These items formed a nine-point scale on which high values indicated a high degree of danger or uncertainty. In the second composite measure (CM-2), the uncertainty questions emphasise an individual's specific differences in their ability to predict product attributes. Questions, 3, 4 and 5 were combined as follows:

$$\text{CM-2} = (3) \times \frac{(4 + 5)}{2}$$

In the third composite risk measure (CM-3), consequences were again represented by ratings of importance (4) and investment (5) as in CM-2. The unpredictability component included the perceived unpredictability of product dependability with repeated use (6), product construction and materials (7),

results of product failure (8), and the degree (9) and kind (10) of goal fulfilment involved. The formula for obtaining CM-3 is seen in Table I.

This has remained one of the most comprehensive measures of product risk to the present day, rivalled only by a number of models which have taken a multi-attribute approach (e.g. Dowling and Staelin, 1994; Greatorex and Mitchell, 1993; Zikmund and Scott, 1977). The number of other studies using a similar approach has been negligible; perhaps because of the amount of data required. It is unfortunate that Deering and Jacoby (1972) did not report on the variation between the measures. Perhaps if they had been able to demonstrate the benefits which might be reaped from collecting such additional information, their expanded measure may have been more extensively used by subsequent researchers. The amount of data required and the lack of information on loss types may have resulted in the decision by many researchers not to use the model.

Multi-attribute models

Zikmund and Scott (1977) began by asking questions such as, do products have certain characteristics which influence the nature of risk consumers perceive? Does the nature of the risk perceived vary by product attribute or the purchase situation? They proposed that some attributes may lead to interpersonal forms of risk, while others may result in more performance-related concerns and their analysis strongly suggested that a product's characteristics do influence the degree of overall perceived risk. When overall risk was divided into its component losses, there was a statistically significant relationship between the set of product characteristics and the set of risk components, in the case of the high and medium-risk products. As the product class becomes more risky, the strength of these relationships (i.e. canonical Rs) increases. This is evidence to suggest that consumers will be more aware of the structure of risk as overall perceived risk increases. Such evidence also forms part of the reasoning for including loss measures in any risk assessment; something which the seemingly more comprehensive Deering and Jacoby (1972) model does not.

Around the same time, Pras and Summer were working on a different problem. Before 1978, the majority of research on multi-attribute models had involved the implicit assumption that the consumer knows with certainty the true values of relevant attributes of the various alternative brands. Pras and Summers' (1978) objectives were to develop and "test" a general procedure for adapting current multi-attribute models to cover "decision making under uncertainty". The basic approach involved the development of "risk-adjusted" brand/attribute ratings based on a consideration of the brand/attribute evaluation process and potential differences in consumers' propensities to accept risk. Only risk associated with uncertainty about the true brand/attribute ratings was considered. Other potential sources of risk, such as uncertainty about the relative importance of various attributes, and others listed earlier were not considered. The proposed risk-adjusted measure (see Table I) proved to be a superior predictor of preferences when compared with its

variants which only considered risk-avoidance behaviour and the mean of each brand/attribute rating distribution (Pras and Summers, 1978).

More recently, Greatedorex and Mitchell (1993) have integrated the multi-attribute and perceived risk ideas and produced a conceptual model of how the two are related. They suggest that the amount of loss is proportional to the degree of mismatch between the required and expected or achieved level on a particular attribute. This amount can be converted into a risk by taking into account the probability of the attribute failing to meet the required level of performance. It will also be affected by the importance of the attribute and the product as well as an individual's tolerance for the loss. The model also incorporates consumers' uncertainty in making any evaluations of required levels, attribute importance, etc., and characterises them as a probability distribution rather than point estimates (see Figure 2). Empirical testing of this proposed model is now a priority for future research.

Considerations and further research

Risk models can sometimes be enhanced by considering the effect of self-confidence on risk perception. Two types of self-confidence have been identified. General self-confidence is the amount of confidence or self-esteem a person has in any situation, while specific self-confidence is the amount of confidence a person has when making a specific purchase decision. Many researchers have found that the relationships between general self-confidence and perceived risk are consistently weaker than those between specific self-

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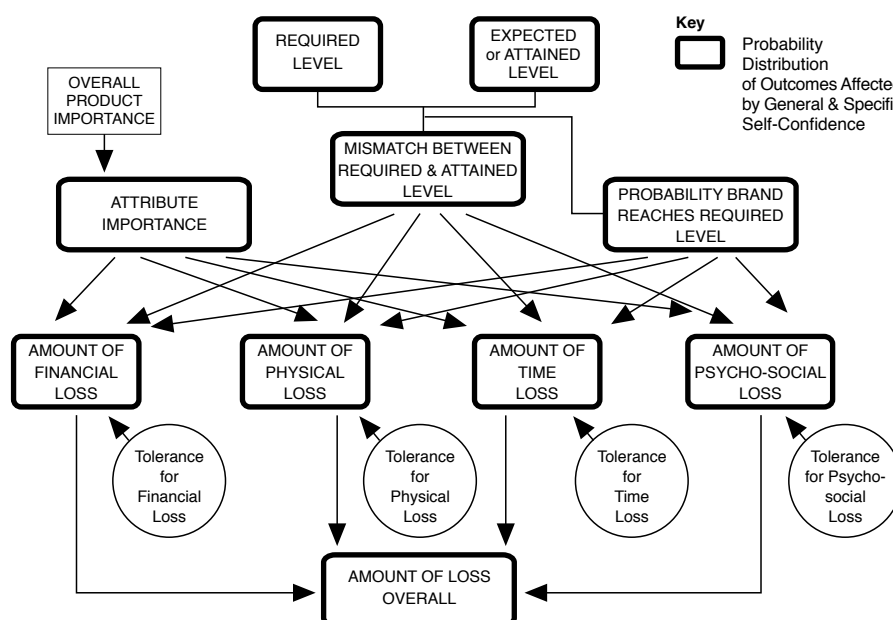


Figure 2.
Flow diagram of risk
processes in a brand
choice decision for a
single attribute

confidence and perceived risk and argue that general self-confidence is essentially a personality measure and can hardly be expected to relate to unique situations to the extent that specific self-confidence can (Cunningham, 1967; Hisrich *et al.*, 1972; Zikmund and Scott, 1973). Nonetheless, statistically significant variations between both general and specific self-confidence and perceived risk were reported by Hisrich *et al.* (1972). The relationship for the former was linear in nature while that between specific self-confidence and perceived risk appears to be asymptotic. As specific self-confidence increases perceived risk decreases, but then levels off. Consumers apparently sense a certain amount of risk in choosing a store and no amount of expertise on their part can remove this uncertainty (Hisrich *et al.*, 1972, p. 438).

In 1979, Churchill suggested a paradigm for developing better measures of marketing constructs and his suggestions give a starting point for an initial evaluation of risk measurement models (Churchill, 1979). The first step is to specify the domain of the construct by delineating what is included in the definition and what is excluded (Churchill, 1979, p. 67). In the perceived risk literature, this has rarely been done to such exacting standards; many authors have simply used Bauer's and Cox's (1967) initial ideas. Churchill (1979) also recommends that the generation of sample items to encapsulate and operationalize the construct should be thorough, involving critical incident technique, focus groups and information from prior research etc. In much of the perceived risk work, this aspect appears to have been treated superficially with very little being reported on where items came from (see Dowling and Staelin, 1994, for an exception).

In 1973 Bettman noted: "Future research might include multiple methods of measuring risk and the other constructs of the model to further examine issues of reliability and validity" (Bettman, 1973, p. 184). Peter (1979, p. 15) in noting that there are many types of perceived risk (e.g. financial, social, etc.) commented that "perhaps a multi-item scale is needed for each type". Finally, Stone and Gronhaug (1993) have recently reiterated: "... multiple measures of these constructs (risk dimensions) are virtually non-existent in the marketing literature". Stone and Winter (1985, p. 10) also acknowledged that comparisons between risk measurements are difficult because different researchers conceptualise risk uniquely and they suggest that this is one of the major reasons for a waning of interest in the concept. The majority of the literature reports unidimensional measures of risk, namely a single statement which either measures overall risk, or the probability component or the consequences component, e.g. how important is psychological risk when purchasing X? In more in-depth studies, these unidimensional measures are sometimes applied to the types of risk, e.g. financial uncertainty or social consequences.

One solution is to measure the risks indirectly through statements generated from in-depth interviews. Instead of asking "what are the social risks involved in the purchase?", several statements could be used to replace the overall concept of social risk, e.g. your superiors will be displeased, or, your relationship with colleagues may be adversely affected. These statements are

more meaningful to respondents and therefore do not require briefing. In addition, by specifying the components of social loss separately, each can be measured individually and the measurement of the construct, e.g. social risk, should be more accurate. Furthermore, with multiple statements measuring the same construct, tests of reliability and validity are possible. This solution also helps overcome the briefing problem associated with trying to explain what is meant by risk to consumers. Kelly's construct elicitation technique could be useful in this respect. Respondents could be given three competing brands and asked how two are similar in the risks they present and different from the third. Sub-scales can be specifically designed to measure one aspect of risk, e.g. financial and could be assessed using measures such as Cronbach's alpha and beta. There are several recent examples of this (Mitchell, 1995; Stone and Gronhaug, 1993).

One must be careful, however, not to blindly adopt measures of reliability which are inappropriate and produce misleading results. For example, perceived risk might be considered as an index or a "formative measure"[2]. For such measures, internal consistency reliability measure is not a valid criterion and one would not necessarily expect internal consistency among the various items. For example, a purchase might involve high social risk, but low financial risk. Thoughtless application of reliability tests would suggest that the perceived risk measure for this type of purchase has unacceptably low reliability, when, in fact, such an assessment is simply inappropriate. Even within a particular perceived risk category, there may be no reason to expect consistency. Thus, a purchase may involve social risk in terms of one's relatives, but not in terms of one's colleagues. This would again cause internal inconsistency scores to be low, even though the measure might be both reliable and valid.

When considering the models presented in this review against the proposed criteria of understanding, prediction, reliability and validity, practicality and useability (see Table I), those models which appear to come out best are the simpler models. While it is acknowledged that a major test of a model's usefulness is its "fitness for purpose" and as such it is unlikely that one model will suit every purpose, the simpler models are likely to fit or be adaptable to more situations than other models. Cunningham's (1967) two-component model, or some modification of it, is a good example of these simpler models and has been one of the most popular models used in the measurement of risk perceptions. There are several reasons why the would-be perceived risk researcher might initially choose to use this model.

First, the two-component model of probability and consequences has been used since its conception in 1967 by many researchers and has a long-standing tradition. If the measure had not proved to be of some worth, it is unlikely that its history would be so long. Second, there are many studies which have employed it over its 30-year history (e.g. Bearden and Mason, 1978; Boze, 1987; Carrol *et al.*, 1986; Dash *et al.*, 1976; Dunn *et al.*, 1986; Greatorex *et al.*, 1992; Guseman, 1981; Hirisch *et al.*, 1972; Hoover *et al.*, 1978; Peter and Ryan, 1976;

Schaninger, 1976; Verhage *et al.*, 1990). With such an extensive use of a similar measure, it seemed more logical to utilise this wealth of data for possible comparison purposes. The two-component model thus affords the greatest degree comparability. Third, Lumpkin and Massey (1983) showed that the two-component model has some degree of convergent and discriminant validity. Also, from a meta-analysis of more than 100 findings, Gemunden (1985) concluded that separate measures of negative consequences and uncertainty allow a better prediction of the amount of information search (a major risk reducer). Fourth, research by Mitchell and Greatedorex (1990, 1993) shows that services are inherently more risky than products and that the major reason for this is the higher levels of uncertainty which are associated with services. They strongly recommend that the uncertainty/probability component should be measured when considering perceived risk in services. Fifth, other risk models, such as those proposed by Deering and Jacoby (1972), Horton (1976) or Pras and Summers (1978) require large amounts of data in order to fully specify the model. Pras and Summers (1978, p. 432) suggest that their procedure is not recommended if the total number of brand/attribute combinations is very large because of the great burden it places on the subject. The two-component risk model suggested is relatively simple to use and easy for respondents to understand. For many data collection techniques, respondent fatigue and difficulty of questionnaire completion are major concerns and preclude the use of the more complex models. Sixth, using a two-component model gives researchers the ability to take multiple measures of risk types (e.g. time, financial, etc.) which is important if a fuller understanding of how risk works is to be achieved. Not many researchers have followed this line (see Mitchell, 1991 and Dowling and Staelin, 1994 as recent examples), despite it having been recommended by Peter (1979).

The basic recommended model can be presented as:

$$\text{Perceived risk} = \sum_n \text{importance of negative consequences} + \text{probability of negative consequences}$$

where n = facets of perceived risk, e.g. time, psychosocial, financial etc.

However, researchers using this basic risk model need to be aware that embedded in the formulation are a number of assumptions. For example, the model implies that each type of adverse consequence, e.g. time, financial, social, psychological, physical, is independent from all others. This assumption may be challenged, since we might expect a faulty product which needs repair always to give some social loss and some time loss. In addition, Jacoby and Kaplan (1972) and Kaplan *et al.* (1974) have found some types of adverse consequences to be highly positively correlated.

A second type of independence assumption relates to the probability and consequences components. The "uncertainty times consequences" orientation has been the perspective for risk since it became an area of research interest to marketers, but it has been criticised by Stone and Winter (1987). The question can be asked: should the two components of risk be judged as independent of

one another, as suggested by the early research of Cunningham (1967), or should they be treated as being related? Bettman (1973), Horton (1976) and Laurent and Kapferer (1985) have found a positive correlation between these two general components of perceived risk. Bettman's correlations ranged from -0.25 to -0.27 , and he concluded that the constructs are not independent. Peter and Ryan (1976) and Bearden and Mason (1979) have found that importance of loss adds little to the predictive validity of the equation, which implies a high correlation between the two components. In psychology, Lanzetta and Driscoll (1968) and Kahnemann and Tversky (1979) provide empirical evidence that subjective probability and utility are not independent. Recent findings by Verhage *et al.* (1990, p. 300) confirm that "a statistically significant association exists between the two components of the perceived-risk measure". One hypothesis which could continue to be tested in future research is that the probability and consequences components of risk are unrelated. Current evidence implies that the two basic components are not distinct and separate constructs.

The formulation also assumes that the two components have equal weighting in the equation. Evidence from Peter and Ryan (1976) and Bearden and Mason (1978; 1979) suggests that the importance of loss or consequences component is less important, while Diamond (1988) provides us with evidence to the contrary. At present, the evidence for rejecting the equality assumption is therefore equivocal and requires further research.

Any set of models which attempts to represent a theory must face tests of its explanatory power. Perceived risk models have had varying success depending on how risky the decision is. Dowling (1986) examined 19 correlation coefficients used to estimate the relationship between perceived risk and product preference (see Arndt, 1967; Bearden and Mason, 1978, 1979; Evans, 1981; Gronhaug, 1975; Peter and Ryan, 1976; Peter and Tarpey, 1975). The average amount of variance explained was 19.4 per cent (standard deviation, 14.3 per cent). Dowling (1986) suggests that being able to explain approximately 20 per cent of the variance in product preference with a single construct should not deter future use of perceived risk by consumer researchers; especially in view of the results of a review of the personality-social psychology literature where Sarason *et al.* (1975) found that situational effects accounted for an average of 12.8 per cent of people's behaviour, personality accounted for an average of 9.4 per cent and demographic factors accounted for an average of only 1.5 per cent of behaviour.

For perceived-risk researchers, the challenge is clear. Current models require much more refinement and development before they can pass the "test of explanation" and from previous research experience in the area, the most difficult concepts to measure accurately are those of social and psychological risk. So intricate, deep-seated and sometimes intimate are the motivations involved in these risks that consumers are sometimes unable to or unwilling to admit their existence even to themselves, let alone to researchers. The importance of not being able to make a proper assessment of these risks cannot

be underestimated not only for its own sake, but also because of the effect these risks have on shaping what can be measured, e.g. time risk, financial risk and physical risk. Since consumers may consciously or subconsciously “pre-select” particular brands for consideration based upon some psychological or social risk dimension, the other risks associated with these brands can be distorted. Even worse, consumers may contrive to misrepresent the performance, financial and time aspects of a purchase in order to justify their ultimate purchase decision based on psychological risks. In these cases, the type and amount of other risks reported will be being governed by the psychosocial risks involved.

In terms of developing and deepening our understanding of how perceived risk works, future research might consider the obvious links between the losses described in the literature and theories of motivation from which these losses are derived. For example the psychological risks of wearing a unacceptably dirty shirt are based on the need for social acceptance. The level and type of need which is being satisfied is likely to be closely related to the main type of risk involved in the purchase. Hitherto, however, the complexity of needs and goals has not been recognised by perceived risk researchers, yet it has a major impact on model development. Mitchell and Hogg (1997) have begun to explore this connection, but much more work is required both conceptually and empirically.

Finally, risk can also be seen to add value to products in some circumstances. For example, in 1965 Berlyne asserted that increasing response conflict can be as important as attempts to reduce conflict, especially in monotonous environments. Persons may engage in “diverse” exploration which introduces new information, “amusement”, “diversion” and “aesthetic experience” (Berlyne, 1965, p. 244). Some authors have noted that repeat buying can account for only a small number of all buying decisions (Frank, 1967) and consumers confronted with a new brand frequently try it without consulting anyone beforehand (Arndt, 1967). Copley and Callom (1971) grouped industrial buyers according to risk perceived across 12 buying situations and found that the relationship of search behaviour to perceived risk varied across the groups; one group behaved as the “Berlyne Curve” would suggest, namely their actions were associated with increased risk. However, only 8 per cent of subjects in their study behaved in this way. Deering and Jacoby (1972) also found risk enhancement occurred at certain risk levels for certain groups. Venkatesan’s (1973) work on novelty seeking is conceptually related and may provide fertile ground for future research into areas of leisure marketing where consumers are regularly attracted to “risky” or dangerous destinations and leisure activities such as ski-ing, hang-gliding, mountaineering, white-water rafting, bungee jumping and other adventurous activities.

Conclusion

The perceived risk concept, with its 38 years of tradition, has stood the test of time and continues to motivate researchers to use its tenet. This is despite, or

perhaps because of, the multiple definitions of the concept. A universally-agreed theoretical or operational definition still eludes marketing academics in the field. Meanwhile, the weight of empirical research has used Cunningham's (1967) two-component model for which there continues to be a good rationale for so doing. The two-component model appears to be the most generally useful and comes out well on the five proposed evaluative criteria of usability, practical implications, prediction, suitability for reliability and validity testing, and developing understanding. However, controversy still exists over whether the two components of probability and consequence should be combined additively or multiplicatively; although the evidence indicates that the additive model is likely to be superior in most cases. Some reservations have been expressed about the independence assumption of the two components and researchers await the results of further work on this question. Until such work is forthcoming, an additive model of probability of loss plus importance of loss is suggested as a working measure.

Good models of perceived risk can only really be judged on what the researcher is attempting to achieve by designing the model. In this respect, each researcher has licence to design objective-specific models which may have very limited general use. Far from this being discouraged, it should be encouraged, but only when other existing models, many of which are presented with this paper, have been evaluated for fitness for purpose.

Notes

1. Interested readers are referred to reviews in the area, e.g. Mitchell (1994, 1995); Mitchell and McGoldrick (1995).
2. A good example of a formative index might be a measure of recreational spending. One might, for instance, measure recreational spending by summing the amount a household spends on attending a movie, eating out, on sporting equipment, for pleasure, travel, etc. The amount spent attending movies would not be expected to, necessarily, correlate with spending on sporting equipment, and internal consistency measures are unsuitable.

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